Digital Electronics Lab Report

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Experiment 3: IC-74181, 4-bit ALU

Objectives:

Verify all 16 arithmetic and 16 logical operations on two 4-bit words, and see the output on a 7-segment LED display or series of bi-coloured LEDs.

Apparatus:

1. IC-74181
2. LED Bar
3. Resistor 470 or 390 - 8
4. Resistor 1k-14
5. Breadboard
6. DIP switches with 4 or more switches-2
7. Connecting wires

8. DC power supply

Theory:

An Arithmetic Logic Unit (ALU) is the core computational block of a digital system. It performs:

1. Arithmetic operations - addition, subtraction, increment, decrement, etc.
2. Logical operations - AND, OR, XOR, NOT, etc.

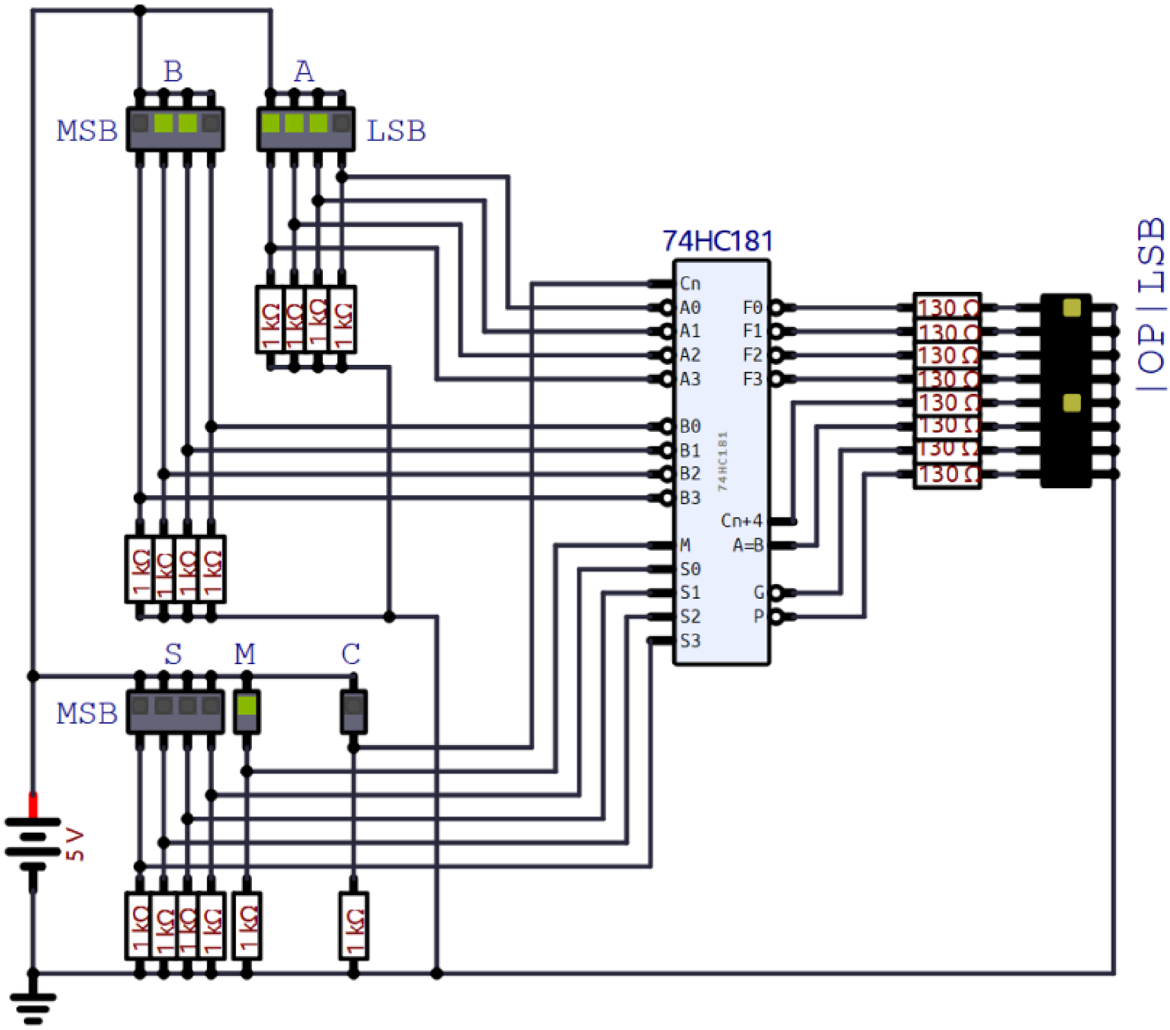
The IC-74181 is one of the most popular ALUs, capable of performing 32 operations on two 4-bit inputs.

Operates on two 4-bit inputs $A_3A_2A_1A_0$, $B_3B_2B_1B_0$ and produces a 4-bit output $F_3F_2F_1F_0$

Function Select Inputs S_0–S_3 are the four pins which select which operation is performed. Total combinations = $2^4 = 16$. Each combination corresponds to a unique operation. Carry input is used only for arithmetic and allows addition with carry and subtraction.

The IC-74181 successfully performs 32 distinct operations (16 arithmetic + 16 logic) on 4-bit binary inputs using control signals. By varying M, S_0–S_3, and C_n, we can verify all operations practically, proving the working of a basic ALU.

Circuit Diagram:

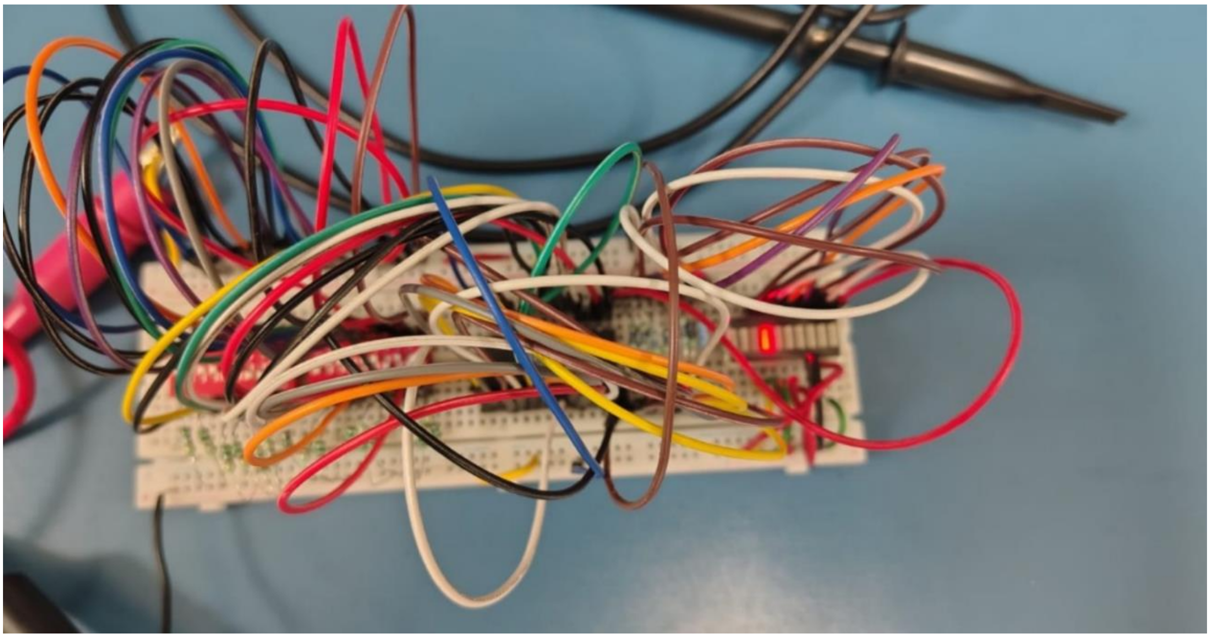


The operands A and B are applied to the ALU. Depending on mode M and function select inputs S_0–S_3, either the arithmetic or logic unit is enabled. The computed result appears at F_0–F_3, while carry and comparison outputs indicate overflow and equality. M is for selecting the mode M=0 is for arithmetic and M=1 is for logical

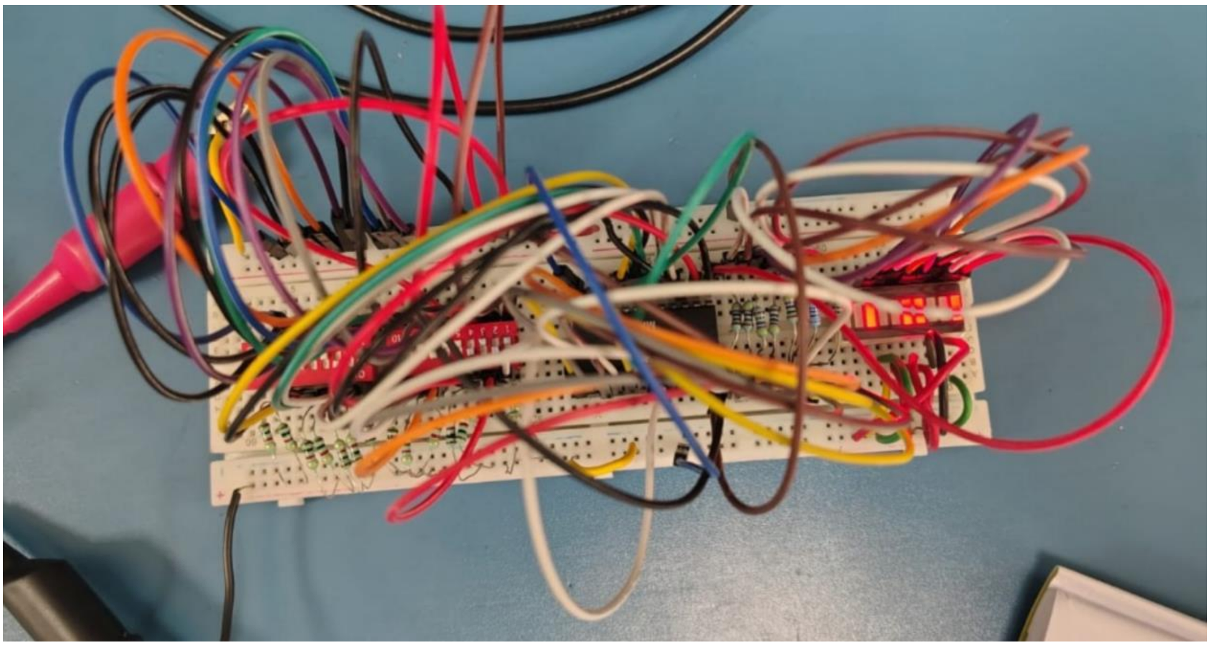
Output/Observation:

Logical : M=H , For all cases A=0101, B=1001

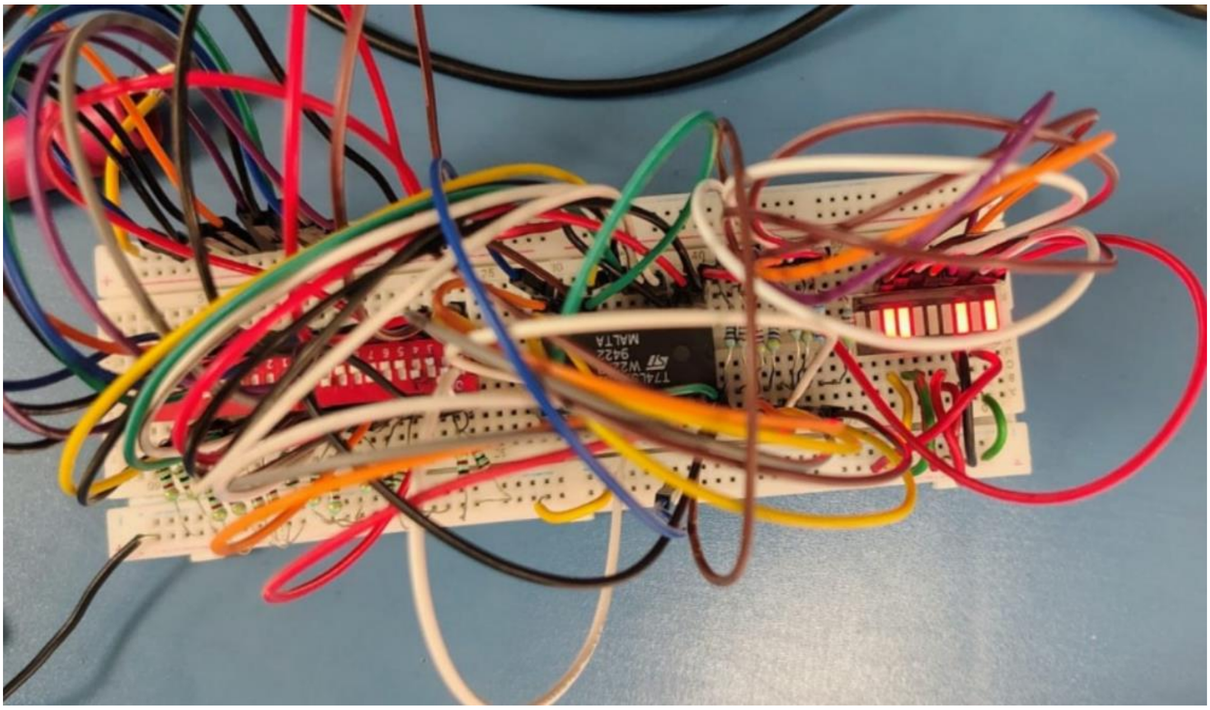
1. LLHH



2. HLHL

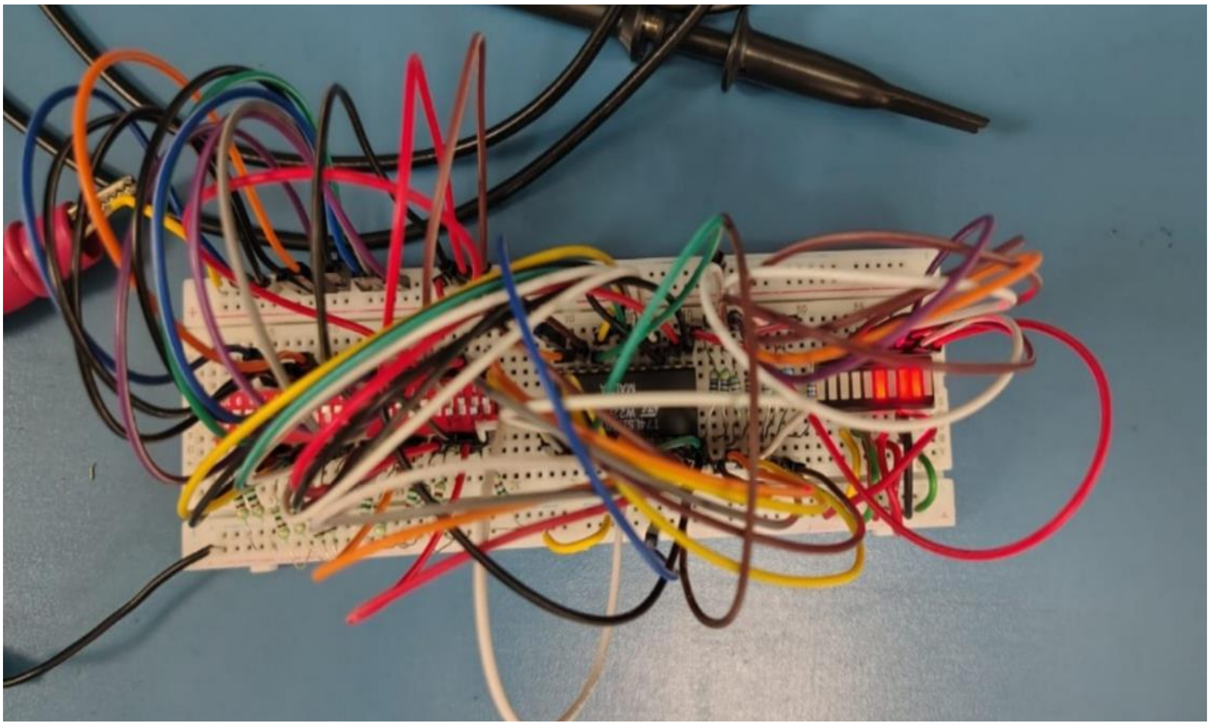


3. HHHH

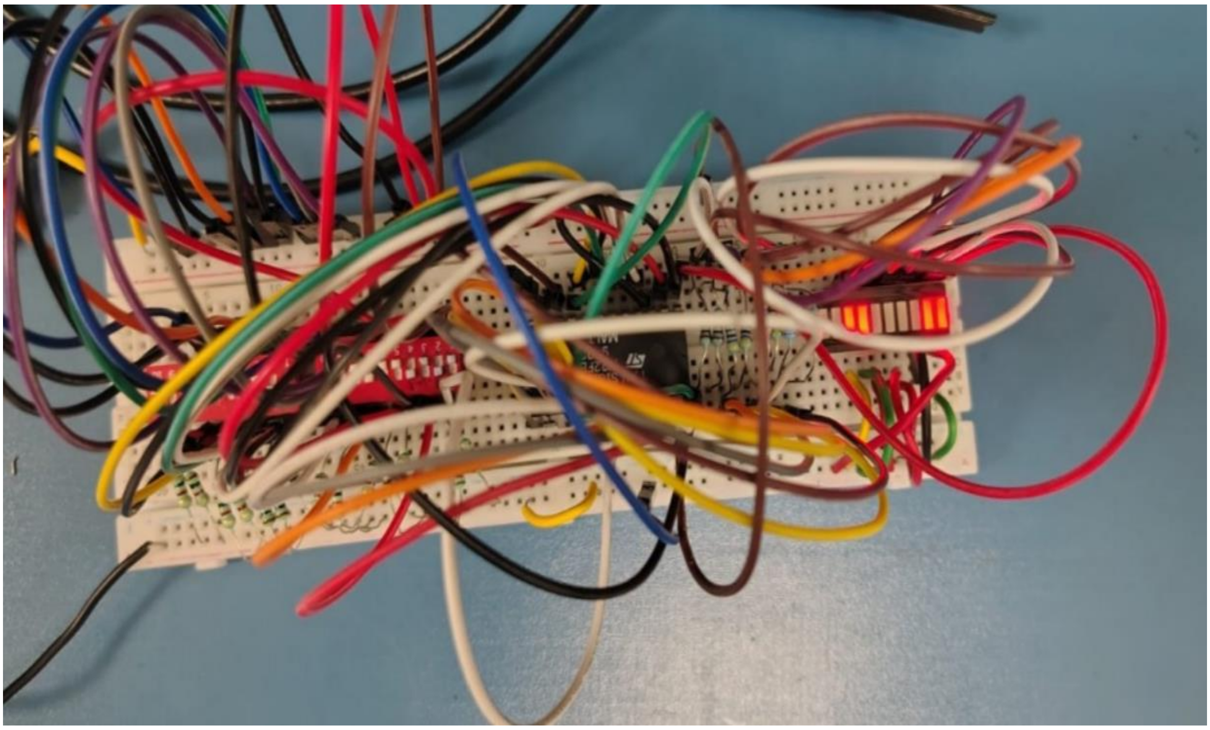


Arithmetic : M=L, Cn=L

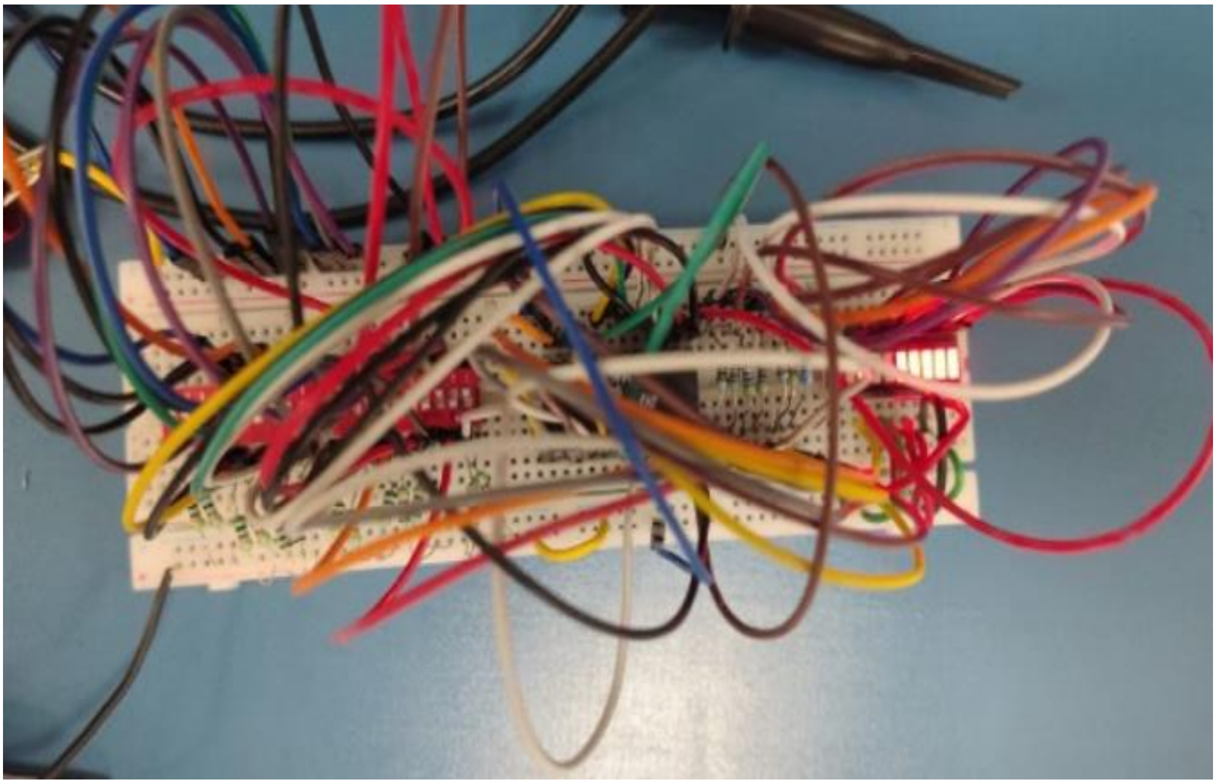
1. LLLL



2. HHLH

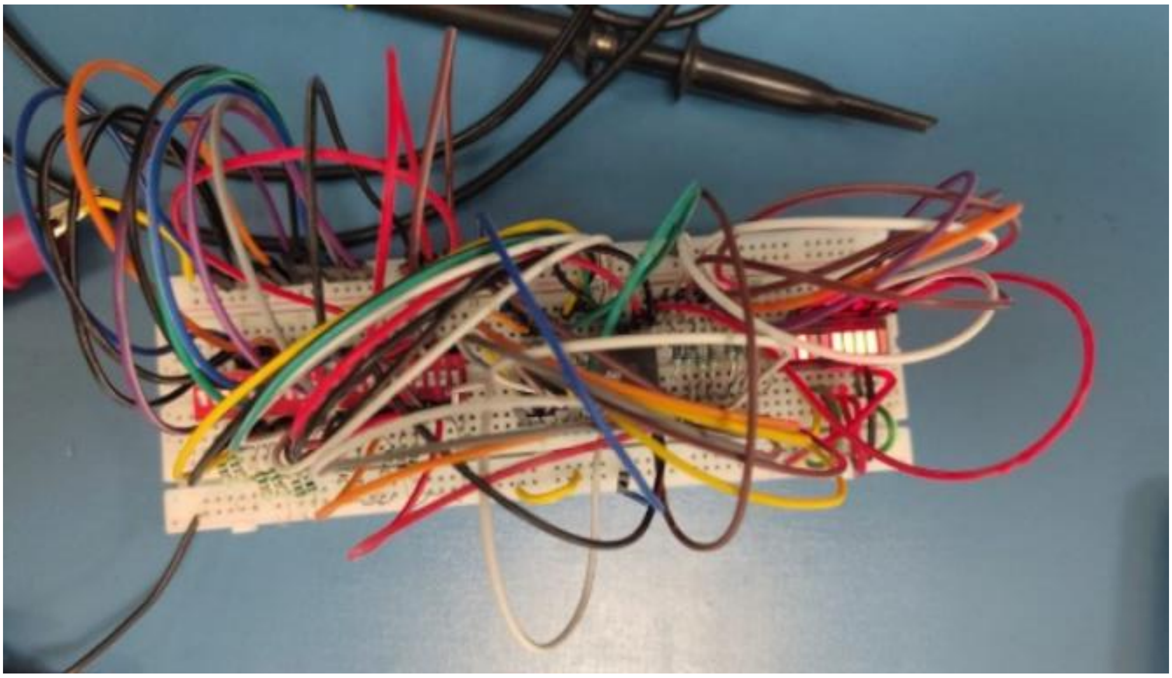


3. HLLH

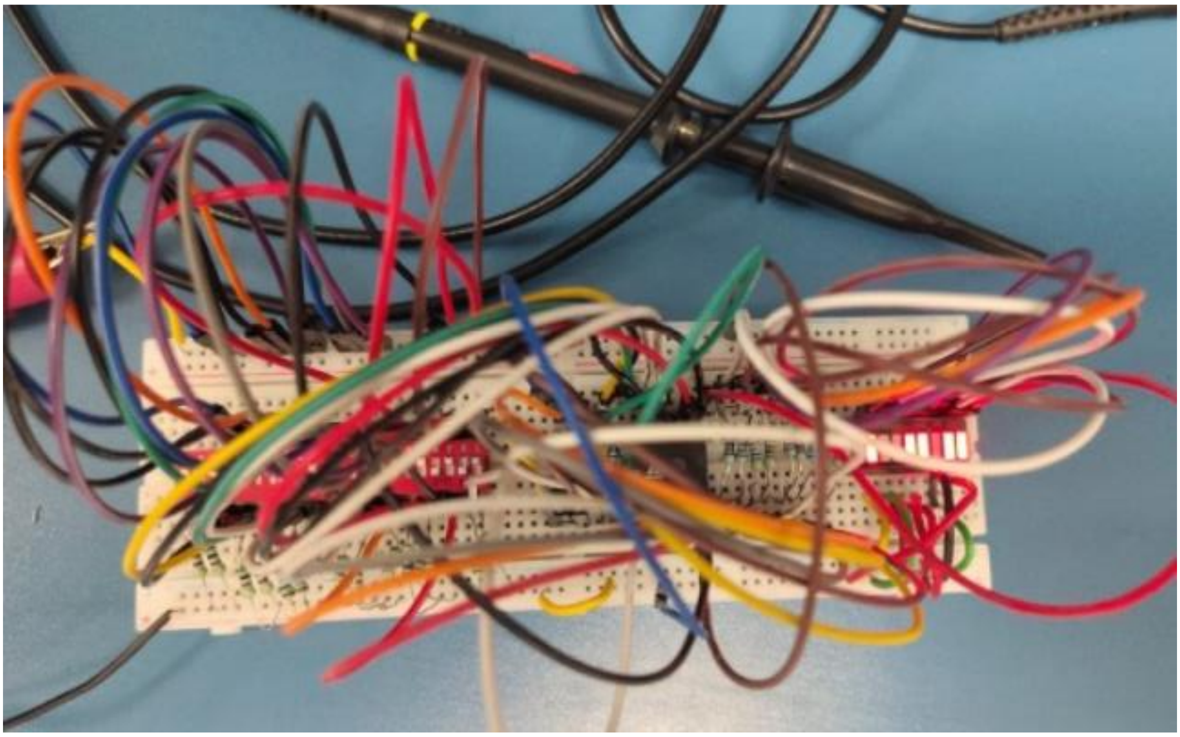


Arithmetic : M=L, Cn=H

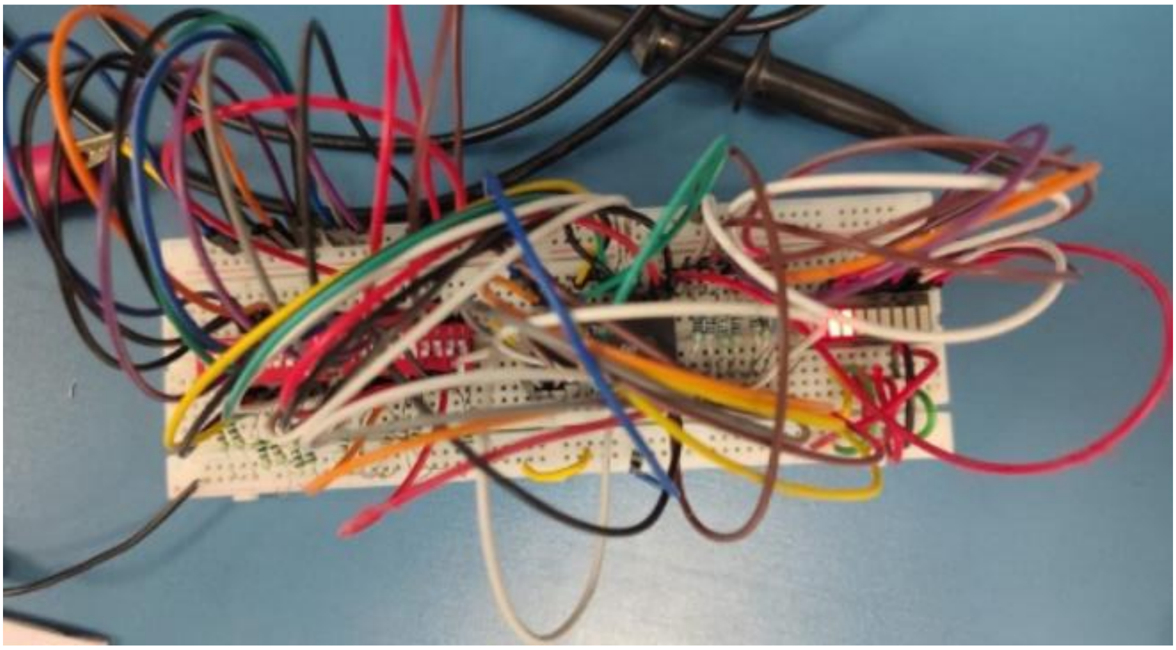
1. HLLH



2. LHHL



3. HLHH



Conclusion:

The IC-74181 (4-bit ALU) was successfully tested and its operation verified. For different select input combinations (S_3–S_0):

Arithmetic mode (M = 0) was verified for both cases of carry input ($C_n = 0$ and $C_n = 1$). Logic mode (M = 1) was verified for the corresponding select inputs. The behaviour under active-high and active-low logic conditions was also verified.

For all test cases, the 4-bit output (F_3–F_0) matched the expected values from the 74181 function table, confirming correct arithmetic and logical operation.